

Grade 7
Unit 8
Lesson 2 Practice

Name _____
Date _____
Period _____

Show your steps of applying the quotient of powers exponent law to evaluate.

1. $\frac{10^5}{10^3}$

2. $\frac{8^4}{8^1}$

3. $\frac{(-6)^6}{(-6)^5}$

4. When dividing two powers that have the same nonzero base, keep the base and _____ the exponents. This is called the quotient of powers exponent law.

Apply the power of quotient exponent law. Do not simplify the fraction.

5. $\left(\frac{-5}{10}\right)^7$

6. Fill in the missing value in the table.

exponent form	4^4	4^3	4^2	4^1	4^0
value	256	64	16	4	

7. Evaluate 9^0

Rewrite the expression by using the laws of exponents. Circle the exponent law used from the list provided. Then, evaluate the exponential expression.

8.

Expression	Exponent Law	Evaluated Expression
$\frac{7.6^5}{7.6^3}$	• Quotient of Powers Law • Power of a Quotient Law • Zero Exponent Law	

9.

Expression	Exponent Law	Evaluated Expression
$\left(\frac{10}{11}\right)^0$	• Quotient of Powers Law • Power of a Quotient Law • Zero Exponent Law	

10.

Expression	Exponent Law	Evaluated Expression
$\left(\frac{0.8}{1.6}\right)^4$	• Quotient of Powers Law • Power of a Quotient Law • Zero Exponent Law	